



Tecnológico de Monterrey

**Reporte Actividad 9:
Conexión a la nube**

Alumnos:

Eduardo Porto Morales - A01027893

Fernando Fuentes Mtz - A01028796

Pedro Mauri Mtz - A01029143

Fecha de entrega:

Junio 4, 2026

Profesor:

Jorge Rodríguez Ruiz

Para crear instancias en la nube:

<http://192.168.133.100/horizon/>

Para conexión por ssh a una instancia:

Shell

```
ssh -i <dirección local donde este guardado el archivo llamado:  
id_nube0ctavo>\id_nube0ctavo <nombre de usuario>@<ip de la instancia>
```

Instancia de Backend: 172.16.80.126

- **Nombre:** backend67
- **Description:** 67 team's backend instance
- **Availability Zone:** Nova.
- **Count:** 1.

Boot Source: debian

Flavor: Tiny

Network: octavoRed

Security group: FrontBack

Configuration:

```
None  
#cloud-config  
ssh_pwauth: true  
preserve_hostname: false  
hostname: 67team-backend  
  
users:  
  - name: team67  
    plain_text_passwd: mauri6porto7fer  
    no_user_group: true  
    groups: sudo, adm  
    sudo: ALL=(ALL) NOPASSWD:ALL  
    shell: /bin/bash  
    ssh_authorized_keys:  
      - id_nube0ctavo.pub  
ssh_deletekeys: false  
  
chpasswd:  
  list: |  
    team67:mauri6porto7fer  
  expire: False
```

Disk partition: Automatic

Conexión de ssh:

Shell

```
ssh -i C:\Users\pedro\Downloads\id_nube0ctavo team67@172.16.80.126
```

Instancia de Frontend: 172.16.80.130

- **Nombre:** frontend67
- **Description:** 67 team's frontend instance
- **Availability Zone:** Nova.
- **Count:** 1.

Boot Source: debian

Flavor: Tiny

Network: octavoRed

Security group: FrontBack

Configuration:

```
None
#cloud-config
ssh_pwauth: true
preserve_hostname: false
hostname: 67team-frontend

users:
  - name: team67
    plain_text_passwd: mauri6porto7fer
    no_user_group: true
    groups: sudo, adm
    sudo: ALL=(ALL) NOPASSWD:ALL
    shell: /bin/bash
    ssh_authorized_keys:
      - id_nube0ctavo.pub
ssh_deletekeys: false

chpasswd:
  list: |
    team67:mauri6porto7fer
  expire: False
```

Disk partition: Automatic

Conexión de ssh:

Shell

```
ssh -i C:\Users\pedro\Downloads\id_nube0ctavo team67@172.16.80.130
```

Contenido de App.js:

None

```
import './App.css';
import { useEffect, useState } from 'react';

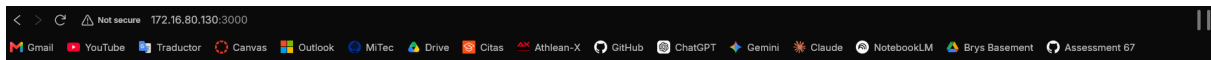
function App() {
  const [data, setData] = useState("");

  useEffect(() => {
    fetch('https://172.16.80.191/api/hola')
      .then(response => response.json())
      .then(data => setData(data.message));
  }, []);

  return (
    <div className="App">
      <h1>Info desde el backend</h1>
      <p>{data}</p>
    </div>
  );
}

export default App;
```

Screen shot del primer frontend trabajando exitosamente:



Equipo 67 - Frontend 1

hola desde el backend

Instancia de Frontend 2: 172.16.80.114

- **Nombre:** frontend76
- **Description:** 67 team's 2nd frontend instance
- **Availability Zone:** Nova.
- **Count:** 1.

Boot Source: debian

Flavor: Tiny

Network: octavoRed

Security group: FrontBack

Configuration:

None

```
#cloud-config
ssh_pwauth: true
preserve_hostname: false
hostname: 67team-frontend

users:
  - name: team67
    plain_text_passwd: mauri6porto7fer
    no_user_group: true
    groups: sudo, adm
    sudo: ALL=(ALL) NOPASSWD:ALL
    shell: /bin/bash
    ssh_authorized_keys:
      - id_nube0ctavo.pub
ssh_deletekeys: false

chpasswd:
  list: |
    team67:mauri6porto7fer
  expire: False
```

Disk partition: Automatic

Conexión de ssh:

Shell

```
ssh -i C:\Users\pedro\Downloads\id_nube0ctavo team67@172.16.80.114
```

Screen shot del segundo frontend trabajando exitosamente:



Equipo 67 - Frontend 2

hola desde el backend

Instancia de Balanceador: 172.16.80.191

- **Nombre:** balancer67
- **Description:** 67 team's balancer instance
- **Availability Zone:** Nova.
- **Count:** 1.

Boot Source: debian

Flavor: Tiny

Network: octavoRed

Security group: Balanceador

Configuration:

```
None
#cloud-config
ssh_pwauth: true
preserve_hostname: false
hostname: 67team-balancer

users:
  - name: team67
    plain_text_passwd: mauri6porto7fer
    no_user_group: true
    groups: sudo, adm
    sudo: ALL=(ALL) NOPASSWD:ALL
    shell: /bin/bash
    ssh_authorized_keys:
      - id_nube0ctavo.pub
ssh_deletekeys: false

chpasswd:
  list: |
    team67:mauri6porto7fer
  expire: False
```

Disk partition: Automatic

Conexión de ssh:

```
Shell
ssh -i C:\Users\pedro\Downloads\id_nube0ctavo team67@172.16.80.191
```

Screenshots de evidencia del balanceador trabajando exitosamente:

```

team67@67team-balancer:~$ sudo nginx -t
sudo systemctl restart nginx
nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
nginx: configuration file /etc/nginx/nginx.conf test is successful
team67@67team-balancer:~$ sudo systemctl status nginx
● nginx.service - A high performance web server and a reverse proxy server
   Loaded: loaded (/lib/systemd/system/nginx.service; enabled; preset: en>
   Active: active (running) since Fri 2026-06-05 03:07:58 UTC; 45s ago
     Docs: man:nginx(8)
  Process: 980 ExecStartPre=/usr/sbin/nginx -t -q -g daemon on; master_pr>
  Process: 981 ExecStart=/usr/sbin/nginx -g daemon on; master_process on;>
 Main PID: 982 (nginx)
    Tasks: 3 (limit: 2314)
   Memory: 2.5M
      CPU: 16ms
   CGroup: /system.slice/nginx.service
           └─982 "nginx: master process /usr/sbin/nginx -g daemon on; mas>
             └─983 "nginx: worker process"
               └─984 "nginx: worker process"

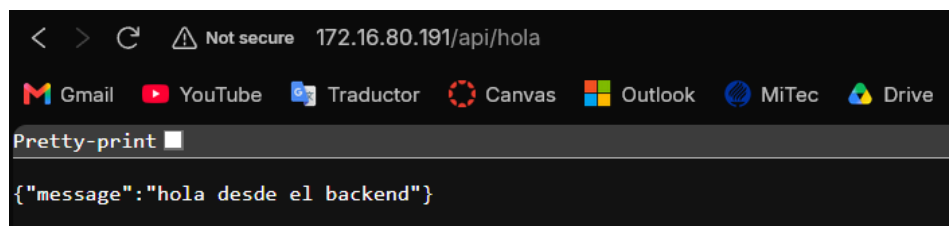
Jun 05 03:07:58 67team-balancer systemd[1]: Starting nginx.service - A high>
Jun 05 03:07:58 67team-balancer systemd[1]: Started nginx.service - A high >
lines 1-17/17 (END)

```

```

http://172.16.80.126:3000/api/hola
{"message":"hola desde el backend"}team67@67team-balancer:~$ curl http://172
.16.80.126:3000/api/hola
team67@67team-balancer:~$ curl -k https://localhost/api/hola
{"message":"hola desde el backend"}team67@67team-balancer:~$ |

```



Screenshot de los 4 servicios ejecutándose:

